ISAN 3348 Design Project II

Now that you have learned a bit about this project from project I, please complete this project with the information provided below.

ABC Company has experienced significant growth over the past five years. They have grown from 20 employees to 348, and they expect to continue to grow at the rate of 12% a year until year 4 when they expect 9% growth.

In response to this growth, the company has leased space in a new business development. This development will provide them with both the space they need immediately as well as the ability to continue to grow without having to relocate in the foreseeable future.

You have been hired to design their data network infrastructure. Your requirement is to identify the host, switch and router configuration. You also need to determine the IP addressing scheme and you need to build a spreadsheet that itemizes all the associated costs associated with your installation.

Current situation:

Total number of employees 348. Total number of hosts, 389.
Number of departments: 8 + Administration

Department host counts*:

Dept.	# Employees	Max throughput Mbps	Annual growth rate	Dept. Name
1	56	1037.88	12%	Marketing
2	64	3492.72	22%	Customer Support
3	29	192.08	8%	Engineering
4	27	21.6	10%	Direct/Corporate Sales
5	9	1228.8	18%	IT
6	97	1299.6	25%	Cold Call Sales Center
7	36	61.25	5%	Accounting
8	18	5.07	3%	Finance
A	12	72.9	8%	Administration

*Employees are distributed proportional to the number of hosts.
Inter department throughput is:

	1	2	3	4	5	6	7	8	A
1	363.258	83.0304	10.3788	0	10.3788	5.1894	0	0	0
2		349.272	34.9272	17.4636	34.9272	0	349.272	0	349.272
3			153.664	3.8416	1.9208	0	0	0	1.9208
4				15.12	0.108	0.108	0.108	0.108	0.108
5					368.64	245.76	122.88	0	122.88
6						129.96	0	0	0
7							6.125	6.125	6.125
8								1.014	1.014
A									18.225

Note: The company uses a VOIP phone system. All throughput not associated with the specified interdepartmental throughput figures can be allocated to the phone system and/or external sources (this would include the internet).

- ***Questions to ponder: Does the growth rate apply to the number of people, the number of hosts, the Max throughput, or all of them?***
- ***What is the difference between the number of employees and the number of hosts?***

Currently the deployment of the departments are spread across 3 buildings. The assignment of departments is based upon current space availability.

Department Location

Building	Floor	Department(s)
1	1	3, 4
1	2	4, 1
1	3	1
2	1	Admin,7
2	2	5, 2
2	3	2
3	1	8, 6
3	2	6
3	3	6

Material and Installation Costs:

Fiber Optic Cable -	$5.00 / ft
Installation cost in building	$20/ft
Installation cost in ground	$100/ft
UTP	$.80 / ft
Installation cost in building	$2/ft
Installation cost in ground	$50/ft
Wired installation under a road	$15,000/road (assumes 90° angle)
HP 5830 layer 3 switch	$8,300 each
Cisco ME 2400 24 port Ethernet switch 10/100	$327 each
Netgear WNDAP620 access point	$434 each

Above is a partial list of items the company has standardized on and the associated pricing they have negotiated with their supplier. You have the ability to choose items other than those provided as long as you provide product specifications, cost, and justification.

Questions to ponder: Is there an easy way to estimate the amount of cable that will be needed?

If you are running wireless, how many access points will be needed?

For this project, which is cheaper: a Wireless system or a Wired system? Explain.

Building Data

Building descriptions

Employees all work in cubicles except for administration.
Support personnel make up 10% of the personnel count for each department and the administration. (For example, if a department has 50 people, then 5 are support.)

support personnel are included in the employee count for a department.

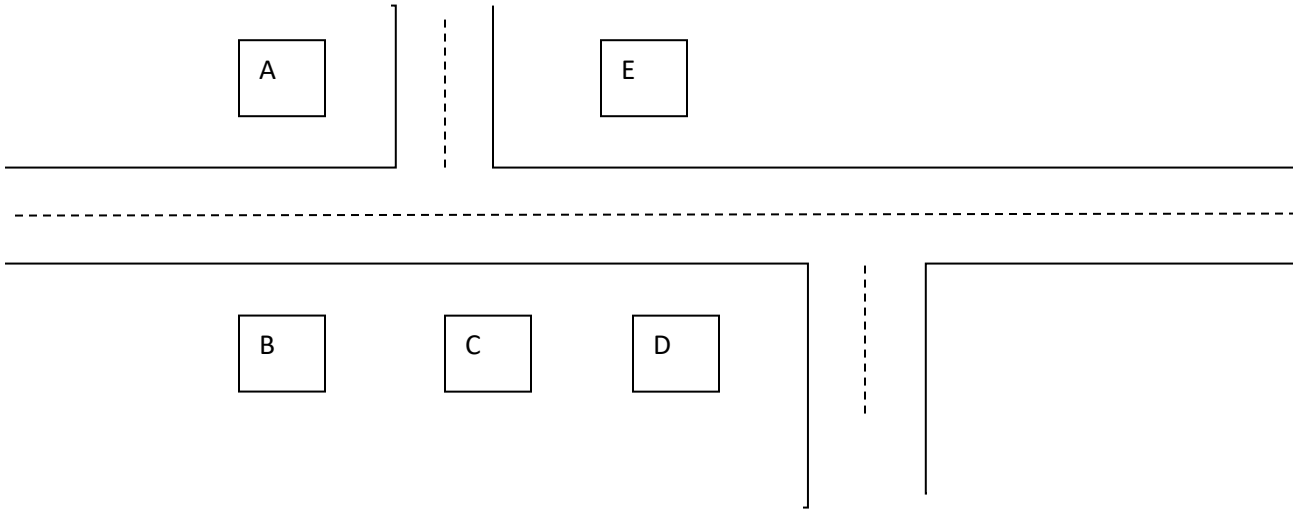
- Employees are assigned cubicles of 122 square feet/person
- Administrators are assigned offices of 240 square feet/person
- Support personnel are assigned cubicles that take up 100 square feet/person
- Allow 30% space for walkways, breakrooms, wiring closets, elevators, restrooms, etc.
- Twenty percent (20%) work from home and don't need dedicated spaces.

Each building is in the shape of a rectangle 75ft by 50ft or 3,750 square feet per floor. There are five floors to each building for 18,750 total square feet.

Elevators are located in the center of each building.

Wiring closets are also located in the same spot on each floor and located on the wall closest to the main road.

Overhead view of location



Buildings are located 100 ft from the street and 200 ft from each other. Building A has been designated by management as the location for the Administrative offices.

The streets are 25ft across.

Special Notes:

- The administrative offices should have a conference room that supports video conferencing.
- Security is a prime concern as the company is entrepreneurial. They want to minimize the ability for theft of their intellectual property as well as their physical assets.
- Security intensive departments (engineering, finance, accounting, administration) should not be located with the call center or customer service.

IP address allocation

Because of the major upgrade to the system that you are working on for the company you have been asked to re-evaluate the company's IP addressing structure.

You have three options:

1. A single IPv4 address at $500 a year.*
2. A range of IP addresses of 147.54.33-70/24 at $15,000 per Class C address
3. IPv6 addresses at $5.00 each

*Note: This option will have additional costs as required by the technology solution for sharing the single IP address.

Deliverables in CANVAS:

A DrawIo/VisualParadigm/ etc. diagram(s) (converted into PDF for final turn in showing the following:* (Use professional software for diagrams)

- Inter-building connectivity with necessary cabling and hardware components taking into account cross building data requirements.
- Intra-building design with locations of all major hardware components such as switches, routers, wireless access points, Internet connectivity, etc.
- IP addresses and sub net mask assignments
- Use icons to display groups of hosts instead of individual ones, and their locations.

An Excel Spreadsheet(s) showing the following:*

- Calculations for determining growth figures for hosts, employees, and throughput.
- Calculations for determining the number of employees and hosts per floor and building
- Department usage calculations
- Cross-Building communications
- Itemized cost sheet for all associated identified costs. Should optimize cost and explain.

Final Report Write-up addressing the following:*

- Assumptions made in creating your solution
- Justifications for choosing the following:
    - Inter-building connectivity structure and media type
    - Intra-building connectivity structure and media type
    - Allocation of departments/employees to buildings
    - Hardware selection and location
    - IP addressing scheme

*Note: the final project report and associated spreadsheets and diagrams should all be designed and presented in formal and professional manner. 10% of the project grade is the presentation of your proposal. Assume you are presenting this to your prospective employer and that your job offer is dependent on this project report.

Grading Criteria

	%Points	
Presentation – professionalism, completeness	10%	
Communication – clarity, spelling, grammar	10%	
Access Layer	10%	
Distribution Layer	10%	
Core Layer	10%	
Data center	10%	
WAN	10%	
Security	10%	
(Inter)Dept Usage Calculations	10%	
Cost calculations	10%	
Total	100%	